Effect of Two Different Training Programs with the Same Workload on Soccer Overhead Throwing Velocity

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A Comparison of Three Training Programs With the Same Workload on Overhead Throwing Velocity With Different Weighted Balls

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Abstract

Van den Tillaar, R and Marques, MC. A comparison of three training programs with same workload on overhead throwing velocity with different weighted balls. *J Strength Cond Res* 25(x): 000–000, 2011—The purpose of this study was to determine if different throwing programs based upon velocity (throwing with a regular sized soccer ball), resistance (throwing with heavy medicine ball), or a combination of both with the same workload would enhance 2-handed overhead throwing velocity with different ball weights. Sixty-eight high-school students (16.5 ± 1.8 years, 57.8 ± 12 kg, 164 ± 9 cm), divided into 3 groups, participated in the study. The training programs were matched on total workload, which resulted in the velocity-training group performing 6 series of 14 reps per session with soccer balls, whereas the resistance-training group performed 3 series of 6 throws with a 3-kg medicine ball, and the combination-training group threw 9 times with a 3-kg medicine ball and 3 series of 14 reps with a soccer ball per session. Throwing velocity with a soccer ball, a 1- and 3-kg medicine ball was tested before and after a training period of 6 weeks with 2 sessions per week. A significant ($p \leq 0.05$) increase in throwing velocity was found after the 6-week training period with the soccer ball (6.9%) and the 1-kg medicine ball (2.8%), but not with the 3-kg medicine ball (−2.5%). In contrast, no group interaction was found with the different balls indicating that velocity, resistance, or a combination as a form of training increased the throwing velocity. Different types of training with the same total workload can increase the throwing velocity in a similar way, which shows that workload is of importance in designing training programs and comparing training with each other. Therefore, those that train high-school soccer players could implement any one of these 3 6-week programs to increase 2-handed overhead soccer throw-in velocity. This could allow the throw-in to be harder or potentially thrown farther if the right trajectory is used.

Key Words specificity, resistance training, velocity training, soccer ball throwing, medicine ball throwing

Introduction

In the performance of overarm throwing, maximum velocity is a very important factor because it is one of the determining factors for throwing distance in javelin throwing, and in team sports, it decreases the time that a goalkeeper has to intercept the ball (18). To improve this variable, training programs based on the principle of overload (by force or velocity of the exercise) can be implemented (17). In sports such as baseball, cricket, and team handball, training programs based on this principle appear to improve throwing velocity (1–5,9,10,12–16,21). However, in many of these studies, resistance training was introduced in accumulation to regular training and compared with control groups that did not get any form of extra training (1,3,5,9,10,13,14). This deficiency makes it difficult to identify which feature of resistance training elicits improved performance: Is it the training form or added training load (8)?

Two studies (8,20) have tried to control the training intensity between experimental groups and a control group by using total workload during training. In ball throwing, momentum of the ball at release (mass ball–m multiplied by the maximal velocity at ball release–v_{rel}) was used to indicate workload (also called impulse), as initial momentum was equal to zero ($\int F dt = \Delta mv = mv_{rel}$) (8). The total workload was calculated with the mass of the throwing object and the number of throws. In both studies (8,20), a resistance-training group was compared with a velocity-training group. In the study of Ettema et al. (8), the resistance-training group had to

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0(0)/1–6

Journal of Strength and Conditioning Research

train resistance in a pulley device mimicking throwing kinematics at 85% of 1 repetition maximum (1RM) with the pulley device. In contrast, van den Tillaar and Marques (20) performed resisted throwing using a 5-kg medicine ball. Both resistance-training groups from the 2 studies were compared with a velocity-training group that conducted throws with a regular weighted handball (0.36 kg) or soccer ball (0.45 kg). It was found that after a training period of 8 (8) and 6 weeks (20), throwing velocity with the regular balls increased with no significant differences between the training groups. Both studies indicated that workload is of major importance when comparing training programs with each other.

Ettema et al. (8) also investigated what the effect of training in a throwing movement at 2 positions (high-velocity position: throwing with regular weighted handballs; high force position: throwing movement in a pulley device) in the force–velocity domain (11) caused on the throwing velocity at the high-velocity end and the low-velocity end of this domain. They observed that training at the high-velocity end of the domain (throwing with regular weighted handballs) also had a significantly positive influence at the low-velocity end of the force–velocity domain (1RM throwing movement at a pulley device). Also, training at the low-velocity end (85% of 1RM) showed positive results in overarm throwing with high velocity. However, what would happen if you combined resistance training and velocity training? Would this type of training have a more positive influence on both sides of the force–velocity domain compared with training on one end of the domain?

In the study of van den Tillaar and Marques (20), 2-hand overhead medicine ball throwing with 5 kg was used as resistance training. However, the effect of training with this weight upon other domains of the force–velocity domain was not measured (except upon throwing with the soccer ball). Thus, they did not investigate if specific heavy resistance training would have a transfer to other parts of the force–velocity domain. The authors suggested that further studies should be conducted on comparing combinations between velocity and resistance training and indicated that combining the 2 training methods (resistance and velocity training) could be useful to limit fatigue related to the total amount of throws.

Therefore, the aim of this study was twofolded: The first aim was to investigate the effect of specific throwing training based on type of resistance (throwing with heavy medicine balls) with type of velocity (throwing with a regular sized soccer balls) or a combination of both on the trained movement when controlled for total workload. Secondly, what is the effect of these training programs on throwing velocity with other throwing weights. It was hypothesized that all groups would improve their throwing velocity because of the additional training with the same workload but that the combination-training group would show the most increase in throwing velocity with the different balls because they trained with both ball weights (specificity principle). A statistically significant difference in throwing velocity between the groups would indicate the influence of the training content.

Methods

Experimental Approach to the Problem

A repeated-measures design with 3 groups (resistance-training group; velocity-training group and a combination-training group) was used to determine the effectiveness of these different specific throwing training programs on throwing velocity with different ball weights during a 6-week training program. A randomized controlled study was conducted in which 3 groups of high-school students, matched on throwing velocity with the soccer ball at the pretest (Table 1), received different training forms (0.45-kg soccer ball throwing, 3-kg medicine ball throwing or a combination of soccer ball and medicine ball throwing) with the same total training workload (i.e., total impulse). The training programs were based on either high resistance, high velocity, or a combination of both. Two-handed soccer ball throwing was used, because most subjects had some experience with this specific throwing technique with medicine and soccer balls. In addition, 2-handed medicine ball and soccer ball throwing mimic the same throwing movements with only a weight difference. This makes it easier to compare the influence of the different training regimens. According to van den Tillaar and Marques (20), the 2-handed overhead throwing technique limits the degrees of freedom that are possible to use

Table 1. Anthropometrics and throwing velocity with the soccer ball of all groups at the pretest.*

Group	Velocity training ($n = 21$)	Resistance training ($n = 21$)	Combination training ($n = 21$)
Weight (kg)	55.5 ± 10.1	60.5 ± 16.6	57.1 ± 7.3
Height (m)	1.63 ± 0.09	1.63 ± 0.10	1.64 ± 0.08
Age (y)	16.3 ± 1.3	16.5 ± 2.1	16.6 ± 1.8
Throwing velocity soccer ball ($m \cdot s^{-1}$)	9.6 ± 1.4	9.7 ± 1.5	9.7 ± 1.7

*Values are given as mean ± *SD*.

(i.e., trivial rotation along the longitudinal axis). Thereby, the performance is less dependent on technique differences between subjects. Because the principal aim of this study was to compare effects of specific resistance-training programs with velocity training on throwing velocity, and not the absolute effect of training regimen, we did not include a nontraining control group (21).

Subjects

Sixty-three (39 women and 24 men) high-school students (age 16.5 $\pm$ 1.8 years, body mass 57.8 $\pm$ 12 kg, height 1.64 $\pm$ 0.09 m) participated in this study. These participants were chosen because none of the participants had extensive experience with this type of training. Furthermore, the participants were not involved in throwing practice besides the throwing training of our study that could influence the results. The subjects were fully informed about the protocol before participating in this study. Informed consent was obtained, before all testing, from all subjects and parents, in accordance with the recommendations of local ethical committee and current ethical standards in sports and exercise research.

Procedures

Before the pretest, the participants were familiarized in 2-handed overhead throwing with different weighted balls in a practice session to avoid a learning effect. Pre and posttests were performed on maximal 2-handed overhead throwing velocity with a soccer ball (circumference 0.68 m; regular weight 0.45 kg), a 1-kg medicine ball (circumference 0.72 m) and a 3-kg medicine ball (circumference 0.78 m). The subjects were tested on the same day of the week and at the same time of the day at the pre and posttest because they followed regular physical education classes. All testing and training were performed from March to April. After a general warm-up of 10 minutes, which included throwing with the different weighted balls to warm up the shoulders, throwing with the different balls was tested. The same procedure as in the study of van den Tillaar and Marques (20) was used. The participant stood with both feet parallel to each other while throwing the balls. All participants started with holding the ball in front of them with both hands. They were instructed to throw the medicine ball as far and fast as possible, while performing the 2-handed overhead medicine ball soccer throw-in. Both feet were kept in contact with the ground at all times during and after the throw and no preliminary steps were allowed. Torso and hip rotation was also prohibited. When a participant did not keep both feet on the ground during the throw, the attempt was not approved, and a new attempt was performed. An expert in throwing, who had years of experience in resistance training with medicine balls controlled this aspect of the study.

Three approved attempts were made with each ball with 1-minute rest between each attempt. The sequence of ball type was randomized for each participant to ensure that fatigue or learning effects did not alter the performance. The maximal throwing velocity was determined using a Doppler radar gun (Sports Radar 3300, Sports Electronics Inc, Washington, D.C., USA.), with a $\pm$0.03 m·s^{-1} accuracy within a field of 10° from the gun. The radar gun was located 1 m behind the participant at ball height during the throw. Only the averages of the 2 best attempts with each ball were used for analysis.

After the initial test, subjects were matched on throwing velocity with the soccer ball and allocated to either the 3-kg medicine ball–throwing group (resistance-training group, n = 21), a 0.45-kg soccer ball–throwing group (velocity-training group, n = 21), or a medicine ball–soccer ball training group (combination-training group, n = 21). All subjects executed the same 2-handed overhead throwing technique while training: standing with both feet on the ground while throwing the ball (either a soccer or medicine ball) as hard as possible to the wall at 3- to 5-m distance (dependent on their throwing skills). Each subject performed these exercises twice a week for 6 consecutive weeks. The participants did not undertake any additional resistance-training activities during the testing or training period. An expert, who had years of experience in resistance training with medicine balls, supervised each training session to ensure that the participants threw correctly and followed the experimental protocol design.

The training workload was calculated by the impulse generated per throwing attempt (8,20). Impulse ($\int F\mathrm{d}t$) was considered a highly relevant measure for resistance training because it measures the total amount of force produced during the throwing movement (8). In ball throwing, momentum of the ball at release (mv_{rel}), measured by the radar gun was used to indicate impulse, because initial momentum was equal to zero ($\int F\mathrm{d}t = \Delta mv = mv_{\mathrm{rel}}$). The comparison of the pretests indicated a mean workload (impulse) for throwing with the 3-kg medicine ball of 20.4 vs. 4.4 N·s for throwing the soccer ball. Thus, to make a comparison with the earlier study on overhead throwing with the same workload of van den Tillaar and Marques (20), the same number of throws with the soccer ball was used as total workload. Hence, the velocity-training group had to perform 84 throws (6 sets of 14 reps) per session with soccer balls (0.45 kg). This resulted in a total workload of approximately 365 N·s. To conduct the same total workload in the other groups, the resistance-training group had to perform 3 series of 6 throws with the 3-kg medicine ball. The workload of the combination-training group was equally divided by the amount of throws of the soccer balls and the throws with the 3-kg medicine ball. This resulted in 9 throws with the 3-kg medicine ball and 3 sets of 14 reps with the soccer ball per training session. A rest period of approximately 3 minutes was used between the series to avoid fatigue.

Statistical Analyses

A 1-way analysis of variance (ANOVA) was performed on the anthropometrics and the throwing velocity of the different training groups at the pretest. To compare the effects of the training protocols, a mixed design 2 (test occasion: pre–post: repeated measures) $\times$ 3 (group: resistance-, velocity-, combination training) ANOVA for each ball was used. To evaluate the performance changes per training group, a 1-way ANOVA for repeated measures (pre–post test) for each ball was performed.

Table 2. Statistical analysis of the effect of resistance-, velocity-, or combination-based training on throwing velocity of the soccer ball and 1- and 3-kg medicine balls of all subject data combined.*†

	Training effect from pre to posttest				Effect between groups		
Ball	% Change (95% CI)	p	Effect size	Statistical power	p	Effect size	Statistical power
Soccer ball	6.9 (4.2–9.5)	<0.001	2.33	0.999	0.554	0.65	0.145
1-kg Medicine ball	2.8 (−0.2 to 5.7)	0.037	0.75	0.557	0.402	0.27	0.203
3-kg Medicine ball	−2.5 (−5.9 to 1.0)	0.155	0.97	0.294	0.288	0.11	0.266

*CI = confidence interval; ANOVA = analysis of variance.
†Overall effect is based on the ANOVA's main training effect and effect between groups on training × group interaction.

To identify differences in training effect between the different balls, a mixed design 3 (change in throwing velocity from pre to posttest with soccer ball, 1-kg medicine ball, 3-kg medicine ball) × 3 (group: resistance-, velocity-, combination training) ANOVA was used.

The test–retest reliability (3 repeats per condition) as indicated by intraclass correlations was 0.91 for throwing velocity with soccer balls, 0.93 and 0.86 for throwing with the 1- and 3-kg medicine ball. The effect size and statistical power are presented in Table 2. The level of significance was set at $p \leq 0.05$.

Results

Pretest data indicated no statistical differences ($p \geq 0.397$) in anthropometrics and throwing velocity between groups (Table 1). When all group data were combined, a significant increase ($p \leq 0.05$) in throwing velocity was found after the 6-week training period with the soccer ball (6.9%) and the 1-kg medicine ball (2.8%) but not with the 3-kg medicine ball (−2.5%; Table 2 and Figure 1). In contrast, no main effect was found between groups with the different balls (Table 2; $p \geq 0.288$).

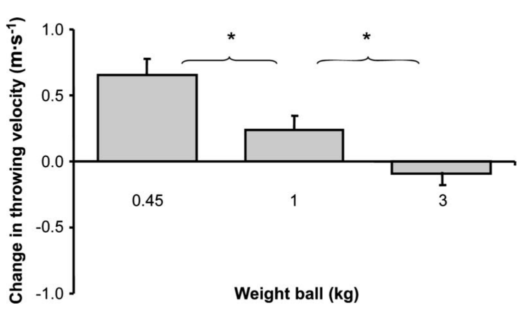

Figure 1. Change in throwing velocity (mean ± *SEM*) with the soccer ball, 1- and 3-kg medicine ball averaged over all participants. *Significant difference ($p \leq 0.05$) in throwing velocity from the pre to the posttest between the 2 balls.

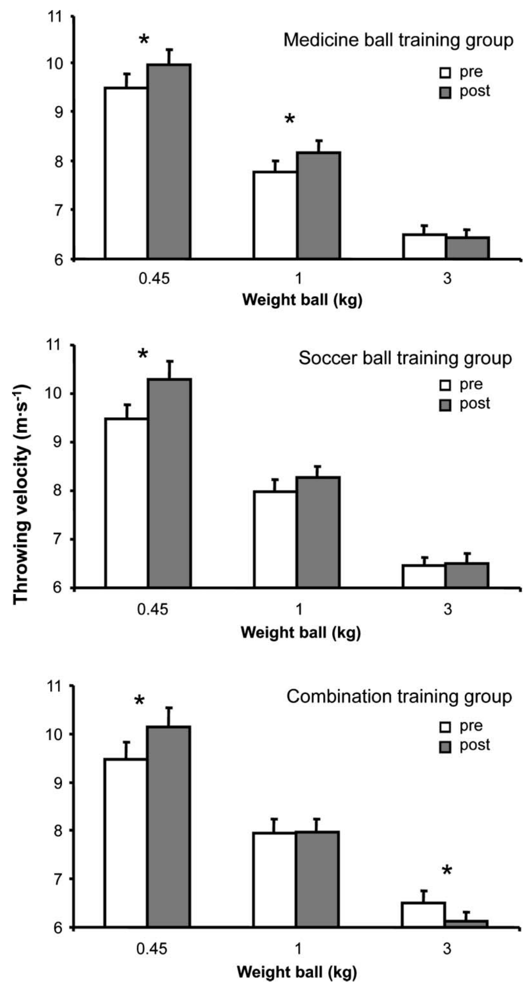

Figure 2. Throwing velocity (mean ± *SEM*) with the soccer ball, 1 and 3-kg medicine ball before and after the training period averaged per training group (resistance- training, velocity-training, and combination-training group) for all participants. *Significant difference ($p \leq 0.05$) in throwing velocity from the pre to the posttest.

When evaluating the performance of each training group, it was found that all groups significantly increased throwing velocity with the soccer ball ($p \leq 0.05$; Figure 2). The medicine ball training group also showed a significant increase in throwing velocity with the 1-kg medicine ball after 6 weeks of training ($p = 0.036$), whereas the combination-training group showed a significant decrease in throwing velocity with the 3-kg medicine ball ($p = 0.037$) at the posttest (Figure 2).

Discussion

All 3 training programs increased the throwing velocity with soccer ball, with no differences between the training groups. A significant increase was also found with the 1-kg medicine ball, whereas no change was observed after 6 weeks of training of throws with the 3-kg medicine ball. The findings of this study are in line with those of van den Tillaar and Marques (20) and Ettema et al. (8) who used similar training programs with the same total workload. The mean throwing velocity with the soccer ball of all subjects (men and women) at the start of the study (9.7 $m \cdot s^{-1}$) was similar to the results reported by van den Tillaar and Marques (20) in women (10.0 $m \cdot s^{-1}$) and much lower than those of the men in that study (13.3 $m \cdot s^{-1}$). These throwing velocities can partially be explained by the height of the subjects as suggested by van den Tillaar and Ettema (19). Van den Tillaar and Ettema (19) found a relationship that the taller the subject is the faster the subject could throw. Furthermore, they found that the increase in height results in an increase in muscle mass, which can explain the increase of throwing velocity. An increase in height also results in an increase in arm leverage, which can have a positive effect on the throwing velocity (19). The women in the van den Tillaar and Marques (20) study were 1.64 m of height, which was the same as the mean height of all subjects (men and women) in this study (Table 1), which resulted in approximately the same maximal throwing velocity as in our study. However, the men in our study were only 1.70 m in height and had a throwing velocity of 10.8 $m \cdot s^{-1}$ with the soccer ball, whereas the men in the study of Tillaar and Marques (20) were 1.78 m in height, and their throwing velocity was 13.3 $m \cdot s^{-1}$. This suggested that the difference in height and thereby also arm length is of importance for the throwing velocity. Furthermore, the mean age of the training group of van den Tillaar and Marques (20) was 22 years, whereas in our study, they were only 16.5 years old. This age difference could influence the strength and experience level of the subjects.

This study showed that the average statistically significant improvement in throwing velocity for all subjects with the soccer ball was 6.9% with an increase of 5, 8.5, and 7.1% for the resistance-training group, velocity-training group, and combination-training group (Table 2). This was also in line with the findings of van den Tillaar and Marques (20) who reported that throwing 84 times with a soccer ball resulted in an increase of 5.1% and that training with other weighted balls also would result in an increase in throwing with a soccer ball. When comparing the increase of throwing velocity with the soccer ball with earlier studies, it shows that in our study the improvements were the highest when looking at groups that trained with other weighted balls or resistances. Training with a 5-kg medicine ball (20) showed an increase of 3.2%, whereas performing throws with 18 kg with a pulley system (8) resulted in only a 1.4% increase in throwing with regular weight balls. These differences are probably caused by the fact that these ball weights stimulate the different parts of the force–velocity curve (9), and improvements may not transfer substantially to throwing with lighter weights. Throwing with 3-kg medicine balls as performed in our study lies closer to the soccer ball weight and has presumably better transfer to throwing velocity with the soccer ball than training with 5-kg medicine balls or with a pulley system. The resistance-training group in our study also showed an increase in throwing velocity with 1-kg medicine balls, which again indicates the possible transfer to a lighter weight (Figure 2).

No change or a significant decrease in throwing velocity was found for all 3 groups when throwing with a 3-kg medicine ball (Figure 2). Van den Tillaar and Marques (20) also reported that training with 5-kg medicine balls did not enhance throwing distance after 6 week of training. An explanation for these results could be that the total workload was not high enough or training period not long enough for increasing throwing velocity with heavier weighted balls. Perhaps a longer training period of 8–10 weeks or an increase in total workload would have altered the results.

It was hypothesized that the group receiving combination training in this study would show the greatest increases because the subjects trained with both ball weights. According to the specificity principle, this would result in increases in both soccer ball and 3-kg medicine ball–throwing velocities. It was also expected that throwing velocity with the 1-kg medicine ball after the 6-week training period would be positively influenced in this group because of a possible transfer from training with heavier and lighter balls (6,7). However, in this group only increased throwing velocity with the soccer ball, no increase with the 1-kg medicine ball and a decrease with the 3-kg medicine ball were observed (Figure 2). Thus, it seems that combining throws with different ball weights with controlled total workload was not enough to enhance performance in throws with heavier balls. An explanation can be that the throwing movement pattern has to be trained sufficiently. In our study, subjects threw at least 18 times with a 3-kg medicine ball or 3 sets of 14 repetitions with a soccer ball per session in an attempt to enhance throwing velocity with a soccer ball. To enhance the throwing velocity of a 3-kg ball, perhaps more throws should have been completed per session before possibly seeing an increase in throwing velocity with these balls.

The total workload between the training groups in this study was equal and elicited a similar increase in throwing velocity with the soccer and 1-kg medicine ball. This indicates that total workload is of importance when designing training programs. As indicated by van den Tillaar and Marques (20), it is not a problem to double the workload in the resistance-training group (36 instead of 18 throws with a 3-kg medicine ball) without losing the quality of throws during the training because of fatigue. This would perhaps also have a positive influence upon the throwing velocity in heavier ball weights. Therefore, it is suggested that further studies be conducted on comparing different total workloads on throwing velocity. Furthermore, researchers need to consider if the number of throws is a determining factor to increase throwing velocity, especially when throwing with heavier balls.

Practical Applications

Three specific throwing training programs with the same total workload, based on velocity, resistance or a combination can increase throwing velocity of a soccer ball by 5–8.5% after 6 weeks of training in a group of high-school students with little soccer ball–throwing experience. This outcome indicates that total workload is of importance for designing throwing programs. Two-handed overhead throwing with 3-kg medicine balls or in combination with soccer balls can also be used as alternative training for improving throwing velocity with soccer balls, if training time is short because the total number of throws is less, and the training can be completed in less time. Therefore, those that train high-school soccer players could implement any one of these 3 6-week programs to increase 2-handed overhead soccer throw-in velocity. This could allow the throw-in to be harder or potentially thrown farther if the right trajectory is used.

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